

The University Stress Scale: Measuring Domains and Extent of Stress in University Students

Helen M Stallman & Cameron P Hurst

To cite this article: Helen M Stallman & Cameron P Hurst (2016) The University Stress Scale: Measuring Domains and Extent of Stress in University Students, Australian Psychologist, 51:2, 128-134, DOI: [10.1111/ap.12127](https://doi.org/10.1111/ap.12127)

To link to this article: <https://doi.org/10.1111/ap.12127>



Published online: 12 Nov 2020.



Submit your article to this journal [↗](#)



Article views: 11669



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 50 View citing articles [↗](#)



ORIGINAL ARTICLE

The University Stress Scale: Measuring Domains and Extent of Stress in University Students

Helen M Stallman¹ and Cameron P Hurst²

¹School of Psychology, Social Work and Social Policy, University of South Australia, and ²School of Public Health, Queensland University of Technology

Objective: Excessive stress and poor coping skills can put university students at risk for mental health problems. The University Stress Scale (USS) provides a measure of both the categories of stress experienced by university students as well as the overall intensity of the stress experienced.

Methods: This study evaluated the psychometric properties of the USS and its clinical utility. Participants were a convenience cross-sectional sample of 2,596 Australian university students.

Results: Exploratory factor analysis provided support for a six-factor model comprising the categories of academic, relationships, equity, parenting, practical, and health. This structure was confirmed using confirmatory factor analysis. The measure had good discriminant and convergent validity and good internal consistency. A cut-off of 13 on the USS indicates students most likely to be experiencing depression and anxiety symptoms as a result of excessive stress. This study used a broad sample of Australian students; however, further replication is needed to confirm the utility of the measure for students in other countries.

Conclusions: The total stress score and the categorical breakdown of stressors makes the USS a useful, brief screening measure for clinicians working with university students.

Key words: college students; factor analysis; psychological distress; stress.

What is already known on this topic

- 1 Excessive perceived stress and poor coping skills are risk factors for mental health problems.
- 2 Stress is commonly measured using one of four approaches: physiological arousal; life events; daily hassles; or symptom checklists.
- 3 Common stress checklists do not provide an overview of the sources of everyday stressor of students that contribute to distress.

What this paper adds

- 1 The University Stress Scale is a psychometrically sound measure of both the domains and extent of stress experienced by university students.
- 2 The USS has particular clinical utility as a screening measure in brief clinical settings, such as counselling and health services, as it provides information that may not be covered in a brief consultation.
- 3 The USS provides an additional approach for assessing stress that uses broad domains rather than specific events or problems.

A degree of arousal is important and necessary for performance; however, too much stress can negatively affect performance and well-being (Easterbrook, 1959; Humphreys & Revelle, 1984; Mills, 1985). The majority of university students report elevated levels of psychological distress (Stallman, 2010), and there is evidence to suggest that this is chronic, with only minor variation between the beginning and end of the academic year (Cooke, Bewick, Barkham, Bradley, & Audin, 2006; Stallman, 2008a; Stallman & Shochet, 2009a). University students report multiple

stressors associated with academic demands; the balancing of study, work, and life; and financial pressures (Ryan, Shochet, & Stallman, 2010). Facing increased stressors, such as the demands of university study, without realistic appraisal or adequate coping skills can lead to significant physical and mental health problems (Hojat, Gonnella, Erdmann, & Vogel, 2003; Patel, Flisher, Hetrick, & McGorry, 2007) and negatively affect academic performance (Andrews & Wilding, 2004; Vaez & Laflamme, 2008).

Excessive levels of stress may account for the high prevalence rates of psychological distress found in university populations (Eisenberg, Gollust, Golberstein, & Hefner, 2007; Stallman, 2008a, 2010; Stallman & Shochet, 2009a). However, it is important to be able to differentiate between levels of stress that may contribute to student success at university and the harmful levels of stress that may lead to poor academic performance, attrition, and/or physical or mental health problems.

Correspondence: Helen M Stallman, School of Psychology, Social Work and Social Policy, University of South Australia, GPO Box 2471, Adelaide, SA 5001, Australia. Email: helen.stallman@unisa.edu.au

Accepted for publication 8 April 2015

doi:10.1111/ap.12127

Stress is commonly measured using one of four approaches: physiological arousal; life events; daily hassles; or symptom checklists. Physiological assessments such as blood pressure, heart rate, respiration, galvanic skin response, or biochemical blood, urine and saliva analyses provide direct, objective, reliable, and quantifiable measures of stress (Baum, Grunberg, & Singer, 1982; Cacioppo, Petty, & Marshall-Goodell, 1985). However, they have several disadvantages including cost, the stressfulness of the actual measures, and the influence of gender, body weight, activity prior to or during measurement, and the consumption of certain substances, such as caffeine, on physiological arousal.

The life events approach to the measurement of stress uses self-report to assess the major events a person has experienced and the stressfulness of each event. The Social Readjustment Rating Scale (SRRS; Holmes & Rahe, 1967) is a widely used scale of life events from which other scales have been developed (e.g., Student Stress Scale; Vickery, 1990). It contains a list of 43 life events (positive and negative) that each has a rating from 100 (most adjustment required) to 11 (least adjustment required). Advantages of the SRRS are that it includes a wide range of events that most people would experience; and the ratings were derived from a broad sample providing an estimate of relative impact of each event. There are, however, several limitations to the life events approach: (a) the reliance on explicit events as the stimuli for stress neglects the fact that stress can result from any part of everyday living and not just major events; (b) the scale does not consider the individual meaning or impact of an event (Cohen, Kamarck, & Mermelstein, 1983; Lazarus & Folkman, 1984); and (c) it has relatively low correlations with illness (Dohrenwend & Dohrenwend, 1984), despite this being one of its main uses (Sarafino, 2008).

Stress is not solely related to major life events but can also relate to everyday problems such as taking an exam, misplacing or losing things, or being caught in traffic. The daily hassle approaches were developed to measure day-to-day unpleasant events (Kanner, Coyne, Schaefer, & Lazarus, 1981). While hassles scores also only have a weak relationship with health, it is a stronger measurement than life events (DeLongis, Coyne, Dakof, Folkman, & Lazarus, 1982). These measures, however, tend to have a large number of items and are time consuming to complete.

Symptom checklists (e.g., Depression Anxiety Stress Scales [DASS]; Lovibond & Lovibond, 1995) assess self-reported, non-specific symptoms of arousal such as difficulty relaxing, nervous arousal, being easily upset/agitated, irritable/over-reactive, or impatient. A major weakness of this approach is that it does not identify stressors that may be the stimuli for the distress and, hence, the target of interventions.

In university clinical settings, such as university health and counseling services, a broad assessment of the various types of stressors may be more useful than an assessment of symptoms of stress or specific events that may act as the stimuli for stress, as it would identify the individual drivers of stress reactions and provide a focus for brief, targeted interventions, and/or appropriate referrals. In combination with a clinical assessment, a self-report measure of stressor categories enables the practitioner to be able to quickly ascertain underlying generic skill

deficits (e.g., problem solving, time management) or specific skill deficits (learning skills, conflict resolution skills), thereby potentially enhancing outcomes in time-limited treatment settings.

The University Stress Scale (USS; Stallman, 2008b) was developed to (a) provide a measure of the severity of stress experienced by university students; (b) allow for the students' subjective experience as to what the stimulus is (by providing only broad category headings rather than specific events or situations) and their cognitive appraisal of the stressor or the intensity of their experienced stress; (c) not restrict respondents' subjective assessment of specific demands and their cognitive appraisal of situations within each category; (d) include not only demands that cause significant stress for most students (e.g., coursework), but also those that may only be experienced by minority groups such as international students or students who are parents; and (e) enable clinicians working in brief counseling settings with students (e.g., student counseling or health services) to be able to quickly identify the breadth of stressors experienced by a student in addition to the presenting problem.

The 21-item measure provides an index of the intensity of an individual's stress experienced in the previous month. The categories of items included were selected through topics identified in focus groups of both students in general and, specifically, international students (e.g., Ryan et al., 2011; Khawaja & Stallman, 2011); a review of self-help topics provided on Australian counseling service websites; and clinical experience. Assessing broad categories rather than specific examples enables the measure to capture anything within a specific category that a student might have perceived as stressful, rather than being limited to specific events or problems identified by the authors. The "other" category enables the student to include additional stressors not covered in the measure.

The aim of this study was to investigate the psychometric properties of the USS, including differences between international and domestic students. In addition, the current study also aimed to provide guidelines on the use of the measure in clinical settings by identifying cut-offs to aid in identifying students whose responses to stress are most likely to be associated with mental health problems.

Method

Participants

Participants were a convenience sample of 2,596 students from five Australian universities. The majority of students were female (72.8%), full time (83.3%), and domestic students (90.2%). Students were aged between 16 and 70 years (mode = 19; median = 22; $M = 25.77$, $SD = 9.12$) and the majority (73.4%) were single. The sample was predominantly Caucasian (67.6%)—the other major ethnic groups were Asian (15.8%) and Middle Eastern (4.4%). Students reporting Aboriginal or Torres Strait Islander ethnicity accounted for .8% of the sample. The majority of students were undergraduate students, with 29.7% being first year students. Students from all faculties were represented with the highest participation rates from Health Sciences (23.4%) and Humanities and Social Sciences (20.6%).

Measures

Student stress

The USS (Stallman, 2008b) is a 21-item screening measure that captures the cognitive appraisal of demands across the range of environmental stressors experienced by students. Students are asked to rate on a 4-point Likert scale ranging from *not at all* to *constantly*, the degree to which each of the categories had caused them stress in the previous month.

Emotional symptoms

The DASS-21 (Lovibond & Lovibond, 1995) was used to assess emotional symptoms. It is a set of three self-report scales designed to measure the negative emotional states of depression, anxiety, and stress. Each scale contains seven items, rated on a 4-point severity frequency scale rating the extent to which symptoms were experienced in the previous week. The depression, anxiety, and stress scales had good internal consistency in this sample ($\alpha = .91$, $\alpha = .82$, and $\alpha = .86$, respectively).

Social support

The Social Support Scale (SSS; Crouchley, Daly, & Molster, 2006) is a five-item questionnaire measuring the perceived amount of social support participants had access to. Responses to each item are on a 4-point scale ranging from 0 to 3 or more people. The SSS has a good internal consistency with this sample ($\alpha = 0.88$).

University connectedness

The University Connectedness Scale (Stallman & Shochet, 2009b) is an 18-item measure of a student's sense of belonging to and support from their university. Students are asked to rate the extent to which each of the experiences is applicable to them on a 7-point Likert scale. The scale had a good internal consistency in this sample ($\alpha = .85$).

Illness

The Patient Health Questionnaire (PHQ; Kroenke, Spitzer, & Williams, 2002) is a self-administered version of the PRIME-MD diagnostic instrument for common mental disorders. The PHQ-15 comprises 15 somatic symptoms from the PHQ, each symptom scored from 0 ("not bothered at all") to 2 ("bothered a lot"). The question regarding menstrual cramps was omitted to enable both males and females to respond to all questions in this study. The measure had good internal consistency in this sample ($\alpha = .82$).

Psychological distress

The Kessler 10 (K10; Kessler et al., 2003) is a brief measure of non-specific psychological distress that has been found to discriminate well between cases and non-cases of depression and anxiety disorders (Furukawa, Kessler, Slade, & Andrews, 2003). Its clinical cut-off in the student population (Stallman & Hurst, 2011) makes it suitable to evaluate the clinical utility of the USS

in university students. The K10 had excellent internal consistency in this sample ($\alpha = .90$).

Procedure

Ethical clearance was obtained from the University of Queensland Human Research Ethics Committee for this study. This study formed part of a larger study into the well-being of university students. All enrolled students at participating Australian universities received an email from the academic registrar in the second week of Semester 2, inviting them to participate in the online survey of student well-being. Upon completion of the survey, participants had the option of entering their name and contact details, that were stored separately to their responses, to be entered into a random draw to win an Apple iPod. Participants were followed up 10 weeks later to complete the survey a second time to investigate the changes in student functioning during the academic semester.

Results

The frequency of student responses for each item on the USS is presented in Table 1. The categories causing at least some stress (any response greater than *not at all*) for most students were *academic/coursework demands*, *study/life balance*, *procrastination*, and *finances and money problems*. Items endorsed least frequently as being stressful were items relating to minority groups within the university including, *sexual orientation issues*, *childcare*, and *language/cultural issues*.

The means and standard deviations for gender, age groups, and student type on the total USS score are shown in Table 2. There was a significant difference in total stress scores for different age groups, with students aged 25 to 34 years reporting higher levels of stress than students aged 35 years and older, although this effect size was very small. There were no significant differences between the older and the youngest groups, or between the youngest and middle groups. Similarly, there was a statistical difference in the total stress scores of males and females, with females reporting higher levels of stress, but again the effect size was small. There was no significant difference between domestic and international students. There was no significant difference between USS scores across universities ($F[4, 2,584] = 1.01$, $p = .42$) indicating that stress as measured by the USS is not related to a particular university.

Exploratory Factor Analysis of USS

Both exploratory and confirmatory factor analyses (CFAs) were performed on the total score of the USS. To avoid overfitting and ensure the external validity of any resulting constructs, different datasets were used for the exploratory and CFAs. One quarter of the participants were randomly selected from the full dataset to use for the exploratory factor analysis (sample 1: $n = 667$; 25%), while the remaining three quarters of participants were retained for the CFA (sample 2: $n = 1,926$; 75%). There were no significant differences between the two samples on gender ($\chi^2 [1, N = 2,593] = .85$, $p = .48$), domestic/international status, $\chi^2 (1, N = 2,593) = .30$, $p = .32$, or age, $t (2,588) = .88$, $p = .38$.

Table 1 Frequency of Responses for Each Item on the University Stress Scale

	Not at all	Sometimes	Frequently	Constantly
	% (n = 2,596)			
1. Academic/coursework demands	4.4	37.9	40.4	17.3
2. Procrastination	12.2	37.0	32.9	32.9
3. University environment	49.4	36.6	10.2	3.8
4. Finances and money problems	15.6	33.0	25.5	26.0
5. Housing/accommodation	59.1	26.1	10.1	4.7
6. Transport	40.2	37.6	15.2	7.0
7. Mental health problems	63.1	25.0	7.9	4.0
8. Physical health problems	49.4	38.3	9.0	3.3
9. Parenting issues	75.0	15.6	6.2	3.1
10. Childcare	88.7	6.9	3.0	1.4
11. Family relationships	42.2	40.5	12.6	4.7
12. Friendships	40.2	48.4	9.1	2.3
13. Romantic relationships	44.0	39.0	12.3	4.7
14. Relationship breakdown	63.7	26.1	6.6	3.6
15. Work	32.2	35.3	22.5	9.9
16. Parental expectations	54.0	27.8	11.6	6.6
17. Study/life balance	9.7	38.3	33.4	18.7
18. Discrimination	79.3	15.5	3.5	1.6
19. Sexual orientation issues	91.5	6.3	1.6	0.6
20. Language/cultural issues	82.8	11.9	3.8	1.5
21. Other demands	83.3	8.9	5.0	2.8

Table 2 Means, Standard Deviations, *F* Values, and Effect Sizes for Demographics on the University Stress Scale Total Score

	<i>n</i>	<i>M</i> (<i>SD</i>)	<i>F</i>	<i>p</i>	<i>d</i>
Total sample	2,593	15.94 (7.81)	—	—	—
Gender					
Males	704	15.14 (7.66)	10.15	.001	.14
Females	1,889	16.24 (7.84)			
Age group					
17–24 years	1,667	15.94 (7.84)	3.04	.048	.17
25–34 years	529	16.51 (7.81)			
35 years+	394	15.23 (7.58)			
Student type					
Domestic	2,345	15.91 (7.75)	.24	.62	.03
International	248	16.17 (8.39)			

Parallel analysis based on principal components analysis (PCA) was performed by generating 1,000 random permutations, as this is one of the most accurate methods of deciding upon the appropriate number of factors to retain (Hayton, Allen, & Scarpello, 2004). To reduce the Type 1 error rate (retaining factors that account for trivial variance), the eigenvalues of the actual dataset (original configuration) were compared with the 95th percentile of the average eigenvalues generated from the random permutations of the actual dataset rather than the mean (Glorfeld, 1995). A decision to interpret only variables with loadings of .3 and above was made a priori.

PCA was used with study sample 1 to establish the number of factors to be retained in the subsequent principal factor analysis (PAF). Six factors had eigenvalues greater than the 95th percentile of eigenvalues of the parallel analysis suggesting a six-

factor structure. The eigenvalues (and percentage of variance explained) associated with the six factors were 4.95 (23.58%), 1.74 (8.30%), 1.51 (7.17%), 1.31 (6.24%), 1.13 (5.38%), and 1.06 (5.03%).

PAF was subsequently fit using oblique (Oblimin) rotation as it would be expected that the USS factors are correlated. The correlation matrix showed numerous correlations greater than .30 suggesting the scale was adequate for factoring. The Bartlett test of sphericity was significant and the Kaiser–Meyer–Olkin measure of sampling adequacy was .82. Inspection of the anti-image correlation matrix revealed that all measures of sampling adequacy were well above the accepted level of .5. Pattern coefficients of items on factors and per cents of variance from the oblique rotation are shown in Table 2. Items are ordered and grouped by size of the pattern coefficient to facilitate interpretation. Interpretative labels are suggested for each factor in a footnote. These are *academic*, *equity*, *parenting*, *relationships*, *practical*, and *health*. The oblique PAF provided low loadings for *parental expectations*, *work*, *university environment*, and *other demands* items, reflecting the heterogeneity of these items comparative to the other items. These items were subsequently excluded from the CFA.

USS Measurement Model

The resulting constructs from PAF were then validated using the CFA. The model used for analysis is shown in Figure 1. The USS was fit to study sample 2 and adequately fit the data. The comparative fit index and the root mean square error of approximation suggested a good fit with values of .935 and .049, respectively, $\chi^2/df = 5.60$; $\chi^2 = 565.40$, $df = 101$, $p < .001$. All items loaded significantly on their respective subscales with

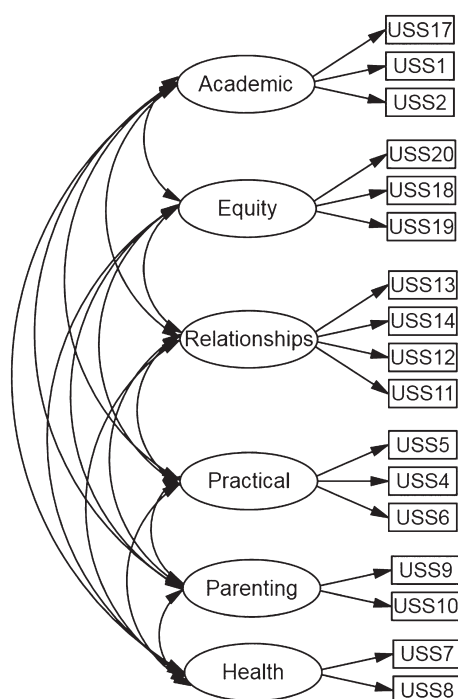


Figure 1 The Hypothesized Fit Model of the University Stress Scale. *Note.* USS Refers to University Stress Scale Item Numbers.

standardized betas ranging from .38 to .96. As shown in Table 3, all subscales were significantly correlated.

Reliability

The overall 21-item scale demonstrated good internal consistency as measured by Cronbach's alpha ($\alpha = .83$). The individual factors showed fair internal consistency with the alphas for academic, equity, parenting, relationships, practical, and health being .62, .63, .69, .73, .64, and .60. Although *parental expectations*, *work*, *university environment*, and *other demands* items did not load significantly on any of the six factors identified by the scale, the deletion of any of them resulted in a reduction in the overall internal consistency of the scale suggesting that individually they contribute to the measurement of student stress.

Twenty five per cent of the sample completed the USS at Time 2. There was no significant difference in Time 1 stress scores between those who did and did not complete the measure at time 2, $t(2,720) \leq 1$, $p = .53$. The intra-class correlation between time 1 and time 2 administrations of the measure was .82 (95% CI .79–.85), showing good test–retest reliability.

Validity

There was a significant correlation between the USS and the stress scale of the DASS, $r = .47$, $p < .001$, indicating good convergent validity. There were also significant correlations with the depression scale, $r = .46$, $p < .001$, and the anxiety scale of the DASS $r = .44$, $p < .001$ and with physical illness as measured by the PHQ, $r = .44$, $p < .001$. The association with physical illness is not significantly attenuated if factor 6 (that includes physical health as a stressor) is removed from the measure,

Table 3 Pattern Coefficients and Per cents of Variance and Covariance for Principal Axis Extraction with Oblimin Rotation on University Stress Scale Items

Item	Variable	F ₁ ^a	F ₂ ^a	F ₃ ^a	F ₄ ^a	F ₅ ^a	F ₆ ^a
17	Study/life balance	.74	-.01	.15	.08	-.11	.04
1	Academic/coursework demands	.67	.01	-.02	-.01	.08	-.08
2	Procrastination	.39	-.07	-.11	-.08	.01	-.03
16	Parental expectations	.28	.20	.17	-.11	-.07	.09
3	University environment	.28	.17	-.04	-.02	-.06	-.10
15	Work	.22	.01	.03	-.11	-.20	-.05
18	Discrimination	.07	.66	.03	.11	-.03	-.09
20	Language/cultural issues	.002	.66	-.004	-.02	-.03	.01
19	Sexual orientation issues	-.10	.40	.03	-.12	-.04	-.03
9	Parenting issues	-.01	-.09	.85	-.08	-.03	-.05
10	Childcare	-.04	.09	.63	.01	.07	-.03
13	Romantic relationships	-.01	-.13	.02	-.84	-.09	.04
14	Relationship breakdown	.01	.11	.03	-.68	.08	-.07
12	Friendships	.14	.13	-.01	-.36	-.14	-.10
11	Family relationships	.11	.02	.25	-.30	-.12	-.09
5	Housing/accommodation	-.10	.06	.04	.03	-.68	-.001
4	Finances and money problems	.15	-.14	-.003	-.02	-.59	-.12
6	Transport	-.02	.10	-.08	-.06	-.52	.01
8	Physical health problems	-.07	.003	.004	-.04	-.04	-.77
7	Mental health problems	.02	.22	-.04	-.14	-.03	-.47
21	Other demands	.08	-.02	.10	.05	-.02	-.24

Note. ^aFactor Labels: F₁ Academic; F₂ Equity; F₃ Parenting F₄ Relationships; F₅ Practical F₆ Health; $\leq .30$ cut-off for inclusion.

$r = .39, p < .001$. The positive associations with these related constructs provide support for the convergent validity with of the USS.

The USS had a negative significant correlation with the SSS, $r = -.27, p < .001$, and a low, but significant, correlation with the University Connectedness Scale, $r = .11, p < .001$. This suggests that while these are somewhat related, they evidently capture different constructs to the USS as indicated by the correlation magnitude.

Clinical Utility

To determine a clinical cut-off on the USS, that is a level of stress that is indicative of psychological problems, a receiver operator characteristic (ROC) curve methodology was used, with greatest sensitivity and specificity on the USS compared to the clinical cut-off recommended on the K10 for university students. Using a cut-off of 20 on the K10 (students ≥ 20) as the predictive value for psychological disorders in university students (Stallman & Hurst, 2011), an ROC curve showed that a cut-off of 13 on the USS gave an area under the curve of .76 (95% CI .75–.78). Given a prevalence of 59.70% of clinical student stress (based on K10), the sensitivity and specificity of the test were 79.15% and 58.95%, respectively. The positive predictive value of the test was 84.20% and negative predictive value of the test was 53.83% on psychological distress.

Discussion

Our aim was to evaluate a brief measure of university student stress that provides a broad assessment of the categories of stressors that are experienced by students and is predictive of mental health problems. Of the 21 items in the USS, 17 combined into a six-factor structure, which was confirmed as adequately fitting the data. The six factors were: *academic, equity, relationships, parenting, practical, and health*. Of the 21 items that were originally considered, 17 were associated to form factors, whereas the remaining four items did not fit into the factor structure. The *work, university environment, and parental expectations* items were sufficiently unique so as to not fit into this structure. The *other demands* item, not surprisingly, did not contribute to any factor, but, nonetheless, strengthens the USS by recording stress that does not relate to the categories specifically mentioned in the scale. The internal consistency of the individual factors is only moderate; however, the complete measure, with the inclusion of the four items that did not load on the factor structure, demonstrates high internal consistency and temporal stability. The strong reliability and validity of the complete measure warrants the inclusion of all items in this instrument.

The measure is significantly correlated with both physical and psychological well-being within this sample. It has a slightly higher correlation with physical illness than the life events ($r = .30$; Dohrenwend & Dohrenwend, 1984) and daily hassles ($r = -.38$; DeLongis et al., 1982) approaches to measuring stress. It is also significantly correlated with a symptom measure of stress and the related constructs of depression and anxiety.

It is important that a measure of stress be able to differentiate between stress and distress, while still displaying the overlap that would be expected between these two constructs. The

results of this study show this relationship between stress and distress, and also highlight that, although related, these are not synonymous constructs. Indeed, some life events can be quite distressing but not stressful and vice versa. The USS is adequately able to make this distinction.

The clinical cut-off score provides some indication of the level of stress that is likely to impact upon student performance, and possibly suggest mental health problems. Given that the majority of students (59.70%) reported stress levels in the clinical range, these findings also support previous research (e.g., Stallman, 2010) that highlighted the need for preventative interventions to enhance students' realistic appraisals and teach the coping mechanisms that are necessary to deal with stressful situations. This will help prevent the challenges that are associated with university study from becoming distressing, thereby placing students at risk for the development of psychological disorders.

Limitations

The large sample size and the lack of differences between students' institution, gender, and age provides some initial support for generalizability of the results from this study. However, the sample for this study was a convenience sample of students studying at Australian universities and may not be representative of all students. Further research is needed to ascertain whether the results in this sample generalize to students in other countries and also in clinical settings.

Conclusion

The USS has utility for clinical work with university students. In services that provide brief interventions to students, such as university counseling and health services, students may only identify a single presenting problem, which is often the one causing the most immediate distress and/or that triggered help-seeking behaviour. One of the challenges facing practitioners in brief treatment settings is becoming aware of the multiple drivers that may be affecting a student's functioning and the relative and perceived severity of those stressors. This is important for understanding the interrelatedness of problems, prioritizing interventions, and identifying referral pathways. The USS, as a brief self-report measure, provides an overview of the severity and specific categories of stress experienced by students, thereby enabling practitioners to better tailor their sessions. This is especially important for those students who are unlikely to attend more than a single session. It has advantages over other measures of stress for this population because of (a) the breadth and relevance of items to both students in general and minority groups; (b) its usefulness in clinical settings; and (c) the clinical cut-off score that discriminates between levels of stress that may be normal and even functional, and those that are likely to be associated with mental health problems.

References

- Andrews, B., & Wilding, J. M. (2004). The relation of depression and anxiety to life-stress and achievement in students. *British Journal of Psychology*, 95(4), 509–521. doi:10.1348/0007126042369802

- Baum, A., Grunberg, N. E., & Singer, J. E. (1982). The use of physiological and neuroendocrinological measurements in the study of stress. *Health Psychology, 1*(3), 217–236. doi:10.1037/0278-6133.1.3.217
- Cacioppo, J. T., Petty, R. E., & Marshall-Goodell, B. (1985). Physical, social, and inferential elements of psychophysiological measurement. In P. Karoly (Ed.), *Measurement strategies in health psychology* (pp. 263–300). New York: Wiley.
- Cohen, S., Kamarck, T., & Mermelstein, R. (1983). A global measure of perceived stress. *Journal of Health and Social Behavior, 24*(4), 385–396.
- Cooke, R., Bewick, B. M., Barkham, M., Bradley, M., & Audin, K. (2006). Measuring, monitoring and managing the psychological well-being of first year university students. *British Journal of Guidance & Counselling, 34*(4), 505–517. doi:10.1080/03069880600942624
- Crouchley, K., Daly, A., & Molster, C. (2006). *An overview of the health and wellbeing of young adults in Western Australia 2002–2005*. Perth: Department of Health.
- DeLongis, A., Coyne, J. C., Dakof, G., Folkman, S., & Lazarus, R. S. (1982). Relationship of daily hassles, uplifts, and major life events to health status. *Health Psychology, 1*(2), 119–136. doi:10.1037/0278-6133.1.2.119
- Dohrenwend, B. S., & Dohrenwend, B. P. (1984). Life stress and illness: Formulation of the issues. In B. S. Dohrenwend & B. P. Dohrenwend (Eds.), *Stressful life events and their contexts* (pp. 1–27). New Brunswick, NJ: Rutgers University Press.
- Easterbrook, J. A. (1959). The effect of motion on cue utilisation, and organization of behavior. *Psychological Review, 66*(3), 183–201. doi:10.1037/h0047707
- Eisenberg, D., Gollust, S. E., Golberstein, E., & Hefner, J. L. (2007). Prevalence and correlates of depression, anxiety, and suicidality among university students. *American Journal of Orthopsychiatry, 77*(4), 534–542. doi:10.1037/0002-9432.77.4.534
- Furukawa, T. A., Kessler, R. C., Slade, T., & Andrews, G. (2003). The performance of the K6 and K10 screening scales for psychological distress in the Australian National Survey of Mental Health and Well-Being. *Psychological Medicine, 33*(2), 357–362. doi:10.1017/S0033291702006700
- Glorfeld, L. W. (1995). An improvement on Horn's parallel analysis methodology for selecting the correct number of factors to retain. *Educational and Psychological Measurement, 55*, 377–393. doi:10.1177/0013164495055003002
- Hayton, J. C., Allen, D. G., & Scarpello, V. (2004). Factor retention decisions in exploratory factor analysis: A tutorial on parallel analysis. *Organizational Research Methods, 7*(2), 191–205. doi:10.1177/1094428104263675
- Hojat, M., Gonnella, J. S., Erdmann, J. B., & Vogel, W. H. (2003). Medical students' cognitive appraisal of stressful life events as related to personality, physical well-being, and academic performance: A longitudinal study. *Personality and Individual Differences, 35*(1), 219–235. doi:10.1016/S0191-8869(02)00186-1
- Holmes, T. H., & Rahe, R. H. (1967). The social readjustment rating scale. *Journal of Psychosomatic Research, 11*(2), 213–218. doi:10.1016/0022-3999(67)90010-4
- Humphreys, M. S., & Revelle, W. (1984). Personality, motivation, and performance: A theory of the relationship between individual differences and information processing. *Psychological Review, 91*(2), 153–184. doi:10.1037/0033-295X.91.2.153
- Kanner, A. D., Coyne, J. C., Schaefer, C., & Lazarus, R. S. (1981). Comparison of two modes of stress measurement: Daily hassles and uplifts versus major life events. *Journal of Behavioral Medicine, 4*(1), 1–39. doi:10.1007/BF00844845
- Kessler, R. C., Barker, P. R., Colpe, L. J., Epstein, J. F., Gfroerer, J. C., Hiripi, E., et al. (2003). Screening for serious mental illness in the general population. *Archives of General Psychiatry, 60*(2), 184–189. doi:10.1002/mpr.310
- Khawaja, N. G., & Stallman, H. M. (2011). Understanding the coping strategies of international students: A qualitative approach. *Australian Journal of Guidance and Counselling, 21*(2), 203–224.
- Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (2002). The PHQ-15: Validity of a new measure for evaluating the severity of somatic symptoms. *Psychosomatic Medicine, 64*(2), 258–266. doi:10.1097/00006842-200203000-00008
- Lazarus, R. S., & Folkman, S. (1984). *Stress, appraisal, and coping*. New York: Springer.
- Lovibond, S. H., & Lovibond, P. F. (1995). *Manual for the depression anxiety stress scales* (2nd ed.). Sydney, NSW, Australia: Psychology Foundation of Australia.
- Mills, I. H. (1985). The neuronal basis of compulsive behavior in anorexia nervosa and related disorders. *Journal of Psychiatric Research, 19*(2/3), 231–235. doi:10.1016/0022-3956(85)90023-8
- Patel, V., Flisher, A. J., Hetrick, S., & McGorry, P. (2007). Mental health of young people: A global public-health challenge. *Lancet, 369*(9569), 1302–1313. doi:10.1016/S0140-6736(07)60368-7
- Ryan, M. L., Shochet, I. M., & Stallman, H. M. (2010). Universal online resilience interventions might engage psychologically distressed university students who are unlikely to seek formal help. *Advances in Mental Health, 9*(1), 73–83.
- Sarafino, E. P. (2008). *Health psychology: Biopsychosocial interactions* (6th ed.). Hoboken, NJ: John Wiley & Sons.
- Stallman, H. M. (2008a). Prevalence of psychological distress in university students: Implications for service delivery. *Australian Family Physician, 37*(8), 673–677.
- Stallman, H. M. (2008b). *University Stress Scale*. Brisbane: Queensland University of Technology.
- Stallman, H. M. (2010). Psychological distress in university students: A comparison with general population data. *Australian Psychologist, 45*(4), 286–294. doi:10.1080/00050067.2010.482109
- Stallman, H. M., & Hurst, C. P. (2011). Factor structure of the Frost Multidimensional Perfectionism Scale in university students. *Australian Psychologist, 46*(4), 229–236. doi:10.1111/j.1742-9544.2010.00010.x
- Stallman, H. M., & Shochet, I. M. (2009a). Prevalence of mental health problems in Australian university health services. *Australian Psychologist, 44*(2), 122–127. doi:10.1080/00050060902733727
- Stallman, H. M., & Shochet, I. M. (2009b). *University connectedness scale*. Brisbane: Queensland University of Technology.
- Vaez, M., & Laflamme, L. (2008). Experienced stress, psychological symptoms, self-rated health and academic achievement: A longitudinal study of Swedish university students. *Social Behavior and Personality, 36*(2), 183–196. doi:10.2224/sbp.2008.36.2.183
- Vickery, D. M. (1990). *Lifeplan: Your own master plan for maintaining health and preventing illness* (2nd ed.). Reston, VA: Center for Corporate Health.